University of Virginia
DS 5004. Applied Reinforcement Learning
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Project Instructions

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Instructions

Students will apply their learning to the goal of completing a substantial reinforcement learning course project. Each student can work individually or in teams of 2-3. This document outlines the necessary components of the final project.

Deliverables

The project has four deliverables:

1. Project Proposal (5% of overall grade)
2. Code (5% of overall grade)
3. Presentation (10% of overall grade)
4. Paper (10% of overall grade)

These are detailed below along with general guidelines and advice. One student can upload each completed item through Canvas. The presentation can be delivered from a student laptop and does not need to be submitted.

1. Project Proposal

The project proposal will outline the project idea and plan. The project must use reinforcement learning. Since the project will be done over several months, it should be substantive and it should be relevant. The proposal can be as short as two paragraphs and as long as two pages.

It will be submitted part-way into the course (the due date will be announced and posted on Canvas). It should include this information at a minimum:

i. Research aim

This is the motivation for the project. Examples:

a. We will develop an RL system to control patient hemoglobin within a defined range
b. We will develop an RL system that avoids dangerous outcomes defined as X

ii. A plan for the data to be used

This should include the source and a brief summary.

iii. Brief discussion of state space, action space, environment, reward function

2. Code Development

To implement the modeling and analysis, it will be necessary to develop code. The code should be in Python.

The code needs to be clearly written and documented in Python scripts (.py format) or Jupyter notebooks (ipynb format). ***Please clearly describe what the code does at the top of each file. Additionally, place the code's "task" in the filename.***

For full credit (5 PTS), the code needs to include these sections in a clean, commented, and comprehensive manner:

i. Data import and preprocessing (1 PT)
Preprocessing includes such tasks as imputing, binning, filtering, outlier treatment, feature engineering, text processing. If this is not necessary, provide justification.

ii. Data splitting / sampling (0.5 PT)
Sampling may not be needed, but splitting is a must. The test set should be left out for evaluation purposes. It should NOT be used in training.

iii. Exploratory data analysis, with at least 2 graphs. Be sure to include these in the paper and presentation (1 PT)

iv. RL algorithms and relevant modeling (1.5PTS)

v. Model evaluation (1 PT)
This should include computation of relevant metrics. For example, if your goal is to keep a patient measurement in a range, what fraction of the time is it maintained by the system in that range? You should demonstrate that the system is working.

3. Project Presentation

One of the exciting things about being a data scientist is that they can drive major change at organizations. As a consequence, they can be called upon to communicate with executives. Strong communication skills (to a technical and non-technical audience) are critical.

In the final weeks of the course, each team will give a group presentation to the class.
Each member must present a portion of the project.

Components of the presentation should include:
i. Executive summary: Discuss the research aims and what you have found
ii. Data summary: explain the state space, action space, etc.
iii. Model / algorithm discussion. As relevant, discuss reward functions, value functions, etc.
iv. Model performance
v. Conclusions and future research

A presentation earning full points will be strong in:
i. Content

ii. Organization / aesthetics
iii. Delivery

4. Project Writeup

The project writeup should include the sections below. The project title should appear in the filename. It could make sense to divide the section writing among teammates; in that case, give the paper a final review for consistency. **The paper should be no more than 7 pages single-spaced.** You can include tables and other artifacts in an appendix; it is not part of the page limit.

Sections:

i. Abstract
Although the abstract appears first, it should be written last. This includes a quick introduction, an overview of what was done, and a summary of findings.
ii. Data and Methods
iii. Results
iv. Conclusions
The conclusions section can include future work, if there was more time.
v. Brief discussion of how each student contributed to the project

5. Project GitHub Repo (recommended but not graded)

Each team member should put their project on their GitHub page.
This should minimally consist of:

i. *README.md* page that summarizes the project: Purpose, major functionality, class methods
ii. Organized code
iii. A *requirements.txt* file listing the required packages

6. Teamwork

Each team member needs to make a substantial contribution and needs to be accountable.
If a teammate issue cannot be resolved within the group, please notify the instructor. Students not contributing meaningfully to the project will not receive an A in the course.

7. Data and Modeling Recommendations

- If you are struggling with massive datasets, drop fields and records that are not needed
- Categorical variables with many levels can often be bucketed effectively. Conduct EDA to understand how best to bucket.
- When testing if the pipeline works properly, try a sample of data to save runtime

8. Final Notes and Advice

- If any issues come up during the course of the project, please reach out so we can address them
- You are encouraged to take on a challenging project, but if it cannot be completed in full and on time, select something simpler
- Practice the presentation beforehand
- Collaborate with others
- Meet face-to-face with your teammates
- Have fun!